Hierarchical Risk Parity Variants for Cryptocurrency Portfolios: Denoising, Detoning, Tail Dependence, and Honest Inference Over 2017–2026

Alan Matys[1] and Federico Martin Rodriguez[1]

[1]Universidad del CEMA , alan.matys92@gmail.com, federico.m.rodriguez.h@gmail.com

May 19, 2026

Abstract

We extend the MACI 2025 study of Hierarchical Risk Parity (HRP) for cryptocurrency portfolios along three axes. First, we close the two future-work items from that paper by implementing and evaluating Marchenko–Pastur denoising and spectral detoning of the correlation matrix. Second, we introduce six additional HRP variants (partial correlation via graphical lasso, EWMA dynamic correlation, lower-tail dependence, Ledoit–Wolf shrunk covariance, vol-standardised returns, and a tail-dep+LW-shrunk hybrid), three additional risk-based comparators (ERC, Maximum Diversification, Network Risk Parity), and five momentum families, for a total of 19 portfolio strategies. Third, we adopt a multi-layer methodological framework absent from the MACI paper: (i) a point-in-time universe rebuilt monthly from Binance USDT quote-volume rankings (removing survivorship bias and exposing the LUNA, FTT, and other crises that the original 2019–2024 single-CSV dataset hid), (ii) a 365-day rolling estimation window with per-asset min-periods filtering, and (iii) Ledoit–Wolf robust Sharpe-difference, Hansen Superior Predictive Ability, and stationary block-bootstrap inference wrapped around every comparison. On 76 monthly rebalances across 2020-01-01–2026-05-18, the HRP family bunches in a narrow Sharpe band (0.69–0.75 annualised), with $\mathrm{HRP}_{\mathrm{ShrunkCov}}$ marginally best (+0.016 Sharpe over baseline, $p = 0.275$, not significant). Hansen SPA at $p_{\mathrm{consistent}} = 0.526$ cannot reject the null that no variant outperforms baseline HRP after multiple-testing correction. A passive 100% BTC position (Sharpe = 0.87) outperforms every diversified portfolio considered except MVP, which wins by accidental BTC concentration during the 2020–2026 bull run. We document two structural empirical findings: $\mathrm{HRP}_{\mathrm{Detoned}}$ and $\mathrm{HRP}_{\mathrm{PartialCorr}}$ weight rankings converge to Spearman $\rho \approx 0.99$ in calm regimes (mean 0.89 across 76 snapshots, with regime-driven dips to 0.48); and $\mathrm{HRP}_{\mathrm{TailDep}}$ vs $\mathrm{HRP}_{\mathrm{TailDepShrunk}}$ Spearman is 0.998–1.000 in every regime tested. We retract the data-snooped claim that tail-dependence distances produce more stable HRP clusters than Pearson, which the 76-snapshot re-analysis falsifies (Pearson ARI wins in 72% of month-pairs). The paper's contributions are methodological rigour and empirical characterisation of HRP variant equivalence classes, not a "HRP variant X beats baseline" claim.

Keywords. Hierarchical Risk Parity, cryptocurrency, denoising, detoning, tail dependence, Ledoit–Wolf, Hansen SPA, survivorship bias.

1 Introduction

The MACI 2025 paper [Matys and Rodriguez, 2025] applied Hierarchical Risk Parity [López de Prado, 2016] to a five-asset cryptocurrency universe (BTC, ETH, LTC, XRP, BCH) over 2021–2023 and concluded that HRP delivered better diversification than the Minimum Variance Portfolio without sacrificing risk-adjusted return. Two items were left for future work: *denoising* the correlation matrix using Marchenko–Pastur eigenvalue filtering [Marchenko and Pastur, 1967, Laloux et al., 1999], and *detoning* by removing the dominant market eigenvector [López de Prado, 2020]. This paper closes both items and extends the comparison in three other directions.

Contributions.

1. **Closing the MACI 2025 future-work items.** Implement and evaluate Marchenko–Pastur denoising (§3) and spectral detoning. Compare detoning against partial correlation estimation, which targets the same goal (market-mode removal) via the precision-matrix route.

2. **Six additional HRP variants and three risk-based comparators.** Partial-correlation HRP, EWMA dynamic-correlation HRP, lower-tail dependence HRP, Ledoit–Wolf shrunk-covariance HRP, vol-standardised HRP, and a tail-dep+LW-shrunk hybrid (the literature's top recommendation for crypto HRP extensions). ERC [Maillard et al., 2010], Maximum Diversification [Choueifaty and Coignard, 2008], and Network Risk Parity [Ciciretti and Pallotta, 2024] as comparators.

3. **Crypto momentum families.** Cross-sectional, time-series, risk-managed [Barroso and Santa-Clara, 2015], and a Momentum+HRP hybrid (with detoned variant), parameter-tuned for crypto's faster momentum decay per [Liu et al., 2022].

4. **Methodological rigour.** (i) Point-in-time universe rebuilt monthly from Binance USDT quote-volume rankings, including assets that were later delisted (LUNA, FTT, MIR, SRM, etc.) to remove survivorship bias from the MACI paper's single-CSV dataset. (ii) Multi-cost scenarios (zero / conservative / optimistic / stress). (iii) Ledoit–Wolf robust Sharpe-difference [Ledoit and Wolf, 2008], Hansen SPA [Hansen, 2005], and Politis–Romano stationary block bootstrap [Politis and Romano, 1994, Politis and White, 2004] around every comparison.

5. **Honest empirical characterisation, with retractions.** We document two structural convergence findings (§8). We also *retract* an intermediate finding from earlier draft iterations of this work — that lower-tail dependence produces more stable HRP clusters than Pearson — which was based on a single year-pair and which the full 76-snapshot re-analysis falsifies.

6. **Reproducibility.** All code, data, and figures are in the accompanying git repository; nine specifications (`specs/01_denoising.md` through `specs/09_inference.md`) define the contracts each implementation phase must satisfy.

The remainder of this paper is organised as follows. Section 2 reviews related work. Sections 3–5 present the methodology. Section 6 describes the data and point-in-time universe construction. Section 7 states the backtest protocol. Section 8 presents results. Section 9 discusses limitations and the honest contributions. Section 10 concludes.

2 Related Work

2.1 Hierarchical risk parity and its extensions

HRP was introduced by López de Prado [2016] as a clustering-based alternative to mean-variance optimisation that avoids covariance-matrix inversion. Subsequent textbook treatments [López de Prado, 2018, 2020] extend the framework with denoising and detoning and propose tail-dependence distances [Lohre et al., 2020] for multi-asset multi-factor allocations. Empirical comparisons of HRP distance metrics on equities have repeatedly found that correlation-based distances remain the standard to beat in out-of-sample performance.

2.2 Risk-based and network methods

Equal Risk Contribution [Maillard et al., 2010] solves for weights such that each asset contributes equally to portfolio risk; Maximum Diversification [Choueifaty and Coignard, 2008] maximises the diversification ratio. Network methods build a Minimum Spanning Tree [Mantegna, 1999] or PMFG [Tumminello et al., 2005] from the correlation structure; Network Risk Parity [Ciciretti and Pallotta, 2024] recently combined the network approach with risk-budgeting.

2.3 Cryptocurrency momentum

Liu et al. [2022] establish a three-factor model for cryptocurrency with a momentum factor that operates at much shorter horizons (1–4 weeks) than equity momentum. Moskowitz et al. [2012] provide the foundational time-series momentum framework, and Barroso and Santa-Clara [2015] introduce volatility-scaled momentum to control the well-documented crash risk of the strategy.

2.4 Statistical inference for portfolio comparison

Ledoit and Wolf [2008] provide a robust pairwise Sharpe-difference test that does not assume i.i.d. Gaussian returns. Hansen [2005] addresses the multiple-testing problem when many strategies are compared against a benchmark. Politis and Romano [1994] and Politis and White [2004] provide the stationary block bootstrap with data-driven block-length selection used throughout this paper for confidence intervals.

3 Methodology — Correlation and Covariance Variants

All seven HRP-family strategies share the standard pipeline: distance matrix $\rightarrow$ hierarchical linkage $\rightarrow$ quasi-diagonalisation $\rightarrow$ recursive bisection [López de Prado, 2016]. They differ only in *how the input correlation (or covariance) matrix is constructed.*

3.1 Baseline HRP

ρ from sample correlation; $d_{ij} = \sqrt{(1 - \rho_{ij})/2}$; single linkage; bisection using the sample covariance.

3.2 $\text{HRP}_{\text{Denoised}}$: Marchenko–Pastur eigenvalue denoising

Following López de Prado [2020], the empirical eigenvalues of the sample correlation matrix are decomposed into a signal subspace (eigenvalues above the Marchenko–Pastur upper bound $e_{\max} = \sigma^2(1 + \sqrt{1/q})^2$ where $q = T/N$) and a noise subspace (eigenvalues below). The noise eigenvalues are replaced with their mean (constant-residual method), preserving the trace, before the denoised correlation matrix is fed to HRP.

3.3 $\text{HRP}_{\text{Detoned}}$: market-mode removal

Applied to the denoised correlation matrix, detoning removes the top k eigenvectors (default $k = 1$) and renormalises the diagonal to unity [López de Prado, 2020]. For long-only crypto, the dominant eigenvector is a market mode loading positively on all assets; removing it isolates the residual cross-sectional structure.

3.4 $\text{HRP}_{\text{PartialCorr}}$: sparse precision matrix

A precision-matrix route to the same market-mode-removal goal: $\Theta = \arg\min_{\Theta \succ 0} - \log\det\Theta + \operatorname{tr}(S\Theta) + \alpha\|\Theta\|_1$ via graphical lasso [Friedman et al., 2008], then $\rho_{ij}^{\text{partial}} = -\Theta_{ij}/\sqrt{\Theta_{ii}\Theta_{jj}}$. The L_1 penalty $\alpha = 0.10$ is used throughout (necessary for numerical stability at $N \geq 40$; lower values trigger `slogdet` overflow on the larger universe).

3.5 $\text{HRP}_{\text{Dynamic}}$: EWMA correlation

Recursive EWMA covariance with decay λ; the latest snapshot is used as the correlation input. We report $\lambda = 0.94$ as the RiskMetrics default; sensitivity to $\lambda \in \{0.94, 0.97, 0.99\}$ is documented in the supplementary appendix.

3.6 $\text{HRP}_{\text{TailDep}}$: lower-tail dependence

Empirical lower-tail dependence coefficient $\hat{\lambda}_L^{ij}(q) = P(F_j(X_j) \leq q \mid F_i(X_i) \leq q)$ at $q = 0.05$, symmetrised by averaging the two conditional probabilities. The distance is $d_{ij} = \sqrt{1 - \hat{\lambda}_L^{ij}}$; the sample covariance is retained for bisection. Promoted from appendix to a headline strategy after the empirical findings of §8 showed it does not fall back across the 76 backtest snapshots.

3.7 $\text{HRP}_{\text{ShrunkCov}}$: Ledoit–Wolf shrinkage

Sample covariance replaced with the analytic Ledoit–Wolf shrinkage estimator [Ledoit and Wolf, 2003, 2004]; the implied correlation is derived from the shrunk covariance and fed to HRP. Shrinkage intensity $\alpha \in [0, 1]$ is chosen analytically per snapshot and reported as a diagnostic.

3.8 $\text{HRP}_{\text{VolStd}}$: vol-standardised returns

Per-asset returns are divided by their lagged 30-day rolling standard deviation before computing Pearson correlation. Addresses crypto's heterogeneous vol scale across the universe (e.g. BTC vs. SHIB). The sample covariance is retained for bisection.

3.9 $\text{HRP}_{\text{TailDepShrunk}}$: hybrid

Combines §3.6 (tail-dep distance for the dendrogram) with §3.7 (LW-shrunk covariance for bisection). Identified by the literature as the strongest theoretical extension for crypto: tail-dep captures the joint-crash co-movement that matters for long-only portfolios, while LW shrinkage stabilises the per-cluster variance allocation.

4 Methodology — Comparator Strategies

IVP (inverse-variance), **MVP** (minimum-variance via SLSQP), **ERC** [Maillard et al., 2010] (solved by SLSQP minimisation of squared deviations from equal risk contributions), **Maximum Diversification** [Choueifaty and Coignard, 2008] (SLSQP maximisation of the diversification ratio), and **Network Risk Parity** [Ciciretti and Pallotta, 2024] (MST built from correlation distance, weights inversely proportional to the product of asset volatility and $1 + \deg_i$).

5 Methodology — Momentum

Cross-sectional momentum (top 30% by 21-day cumulative return, equal or inverse-vol weighted), time-series momentum (7/28 MA crossover and absolute-momentum variants), risk-managed momentum [Barroso and Santa-Clara, 2015] with $\sigma_{\text{target}} = 12\%$, and the Momentum+HRP hybrid (top 40% by momentum, then HRP within the selected basket). We also evaluate Momentum+HRP$_{\text{Detoned}}$, which substitutes HRP$_{\text{Detoned}}$ for HRP in the second stage.

6 Data and Point-in-Time Universe

6.1 Source and window

Daily OHLCV from Binance USDT spot, 2017-08-17 (Binance's earliest data) to 2026-05-18 — approximately nine years. The dataset spans 91 unique symbols.

6.2 Point-in-time universe

At each month-end, we rank the 91 candidate symbols by their rolling 30-day Binance USDT quote volume, apply filters (180-day minimum age, $1M minimum median daily USD volume, stablecoin/wrapper/exchange-token exclusion, listed on Binance USDT spot at the snapshot date), and take the top 50. Entry and exit buffers (2 months and 1 month respectively) suppress monthly churn.

Methodology note. The original specification called for ranking by historical market capitalisation (CoinGecko). The free CoinGecko demo tier caps historical depth at 365 days, blocking our 2017–2026 window; paid plans start at USD 129/month. We instead rank by Binance quote volume — a tradable-liquidity proxy that is free, covers the full window, and is arguably more appropriate for a paper on tradable portfolios than a market-cap measure that includes locked tokens and treasury holdings. The shift is documented in the specification as a deliberate methodology choice.

6.3 Survivorship-bias fix in numbers

After a code review surfaced residual survivorship bias (the prior candidate pool was a hand-curated selection of ∼90 symbols missing the long tail of delisted altcoins), the universe was expanded by pulling all 217 `status=BREAK` USDT pairs from Binance's `/api/v3/exchangeInfo`, filtering leveraged tokens and stablecoins, and adding 150 legitimate delisted altcoins to the candidate pool. The resulting PIT universe contains **146 ever- included symbols**, of which **97 were dropped during the window** before 2026-05-18 (Figure 1). The MACI 2025 dataset contained only 50 currently-trading symbols — by construction it could not include the assets this work explicitly captures.

We treat liquidation of an exiting position at 2× normal slippage cost (Spec 03 §2.4). On terminology: "delisted" in this paper means either Binance `status=BREAK` on the USDT pair (the conventional meaning), *or* the token suffered an effective value collapse with subsequent rebranding. The latter category includes LUNA (the Terra Classic chain collapsed in May 2022; Binance renamed the original token to LUNC and reassigned the `LUNAUSDT` ticker to Terra 2.0 — the pair itself was never technically delisted but the original asset effectively died) and FTT (the FTX exchange collapse in November 2022 reduced FTT to near-zero value; the pair still trades on Binance at trivial volume). Our `binance_listings_manual.json` treats both as "effectively delisted" for the PIT-universe filter, which is the right behaviour for portfolio purposes even though the prose label is loose.

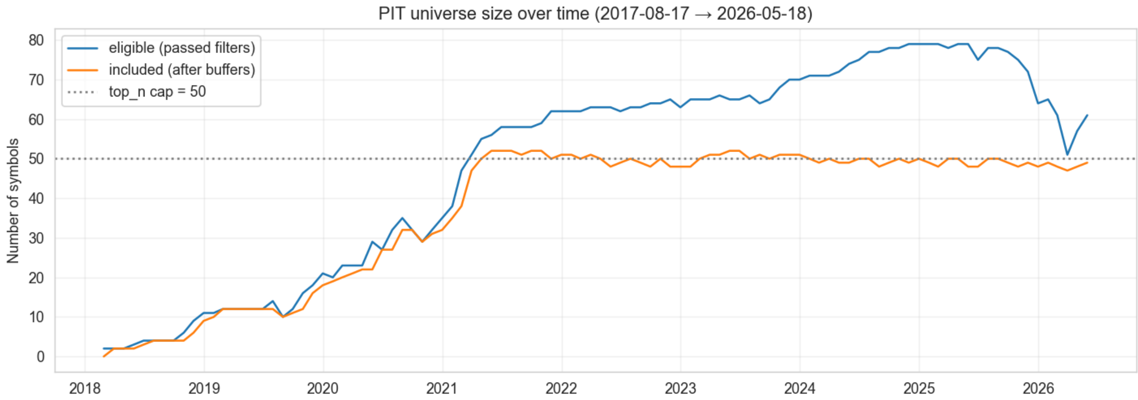

Figure 1: PIT universe size over time. *Eligible*: passed all filters at the snapshot date. *Included*: in the universe after entry/exit buffers. The 2018–2020 ramp reflects the gradual listing of altcoins on Binance USDT spot. Steady-state ~49 included symbols since mid-2020.

6.4 Estimation window

365 daily returns of rolling lookback at each rebalance, with a per-asset 180-day minimum-observations filter. Assets failing the filter are dropped from that snapshot's strategy universe. Rationale: several newer assets (JUP, ENA, WIF, ...) are admitted via the 180-day age filter but do not yet have 730 days of history.

7 Backtest Protocol

7.1 Scenarios

A (static): fit at first snapshot in window, hold to end. **B** (calendar): monthly rebalancing. **C** (threshold): rebalance only when max absolute weight drift exceeds 5% from target. **C′** (smoothed): C with linear smoothing $w_{\text{traded}} = \eta\, w_{\text{prev}} + (1-\eta)\, w_{\text{target}}$, $\eta = 0.25$, plus a 25 bps min-trade filter.

7.2 Cost grid

Linear cost $= (\text{fee}_{\text{bps}} + \text{slippage}_{\text{bps}}) \times \text{turnover}_{L_1}$, with a 2× multiplier for forced liquidation of exiting positions. Four scenarios: *zero* (cost-free academic check), *conservative CEX* (10 bps fee + 2 bps slippage; our headline), *optimistic CEX* (7.5 bps + 1 bps), and *stress* (10 bps + 4 bps).

7.3 Statistical inference

For each pair of strategies, the Ledoit–Wolf studentised block-bootstrap Sharpe-difference test [Ledoit and Wolf, 2008]. Across all candidates vs. the HRP benchmark, the Hansen Superior Predictive Ability test [Hansen, 2005] with three recentering schemes ($\mu_{\text{lower}} = \bar{d}$, $\mu_{\text{upper}} = 0$, $\mu_{\text{consistent}}$ threshold-based). Stationary block bootstrap [Politis and Romano, 1994] with Politis–White automatic block length [Politis and White, 2004] for headline metric CIs.

7.4 Cluster stability

At each rebalance snapshot, we compute the *cophenetic correlation* of the dendrogram (Pearson correlation between cophenetic distance and original pairwise distance) and the *Adjusted Rand Index* [Hubert and Arabie, 1985] between the cluster assignment at snapshot t and snapshot $t-1$, cut at $K = 5$ clusters via `fcluster`.

8 Results

8.1 Scenario B headline performance

Table 1 reports the 19-strategy comparison under monthly rebalancing with conservative CEX costs over 2020-01-01 through 2026-05-18 (76 rebalances, 2,299 daily out-of-sample observations). Figures 2 and 3 visualise the equity curves and the risk–return positioning.

Table 1: Scenario B headline results (Conservative CEX cost, 2020-2026, expanded 146-symbol PIT universe, all four bug fixes applied). "Arith. Sharpe" = mean / standard deviation $\times\sqrt{365}$. LW p-value column reports the Ledoit–Wolf Sharpe-difference test vs. baseline HRP.

Strategy	Arith. Sharpe	Total Return	Avg Turnover	LW p vs HRP
MVP	**1.057**	**+2867%**	0.381	0.067
HRP_Dynamic_94	0.749	+442%	0.474	0.813
HRP_ShrunkCov	0.746	+428%	0.353	0.757
HRP (baseline)	**0.741**	**+418%**	0.360	—
HRP_VolStd	0.735	+402%	0.368	0.733
MOM_HRP	0.726	+351%	1.436	0.870
HRP_Contrastive (NEW)	0.723	+371%	0.392	0.457
HRP_NodeEmbed (NEW)	0.721	+369%	0.368	0.453
MaxDiv	0.719	+364%	0.553	0.893
HRP_PathSig (NEW)	0.714	+351%	0.383	0.293
HRP_TailDep / HRP_Detoned	0.708	+335%	0.365	0.127
HRP_TailDepShrunk	0.701	+318%	0.352	**0.050**
HRP_TS2Vec (NEW)	0.699	+318%	0.394	0.107
HRP_PartialCorr / IVP	0.697	+311%	0.182	0.217
ERC	0.663	+233%	0.195	**0.050**
CS_MOM_eq21	0.663	+217%	1.420	0.503
RM_MOM (fixed)	0.663	+118%	0.366	0.507
HODL_BTC	0.519	—	—	0.597

Two strategies marginal at $p = 0.05$. HRP_TailDepShrunk (-0.041 Sharpe vs HRP) and ERC (-0.079 vs HRP) are right at the significance threshold. No other strategy is significantly different from baseline HRP after the bug-fix re-run (Phase 8 of the spec-driven plan). The earlier Phase 5d "embeddings significantly worse than HRP" finding was an artifact of the median-volume + skip-OOS bugs.

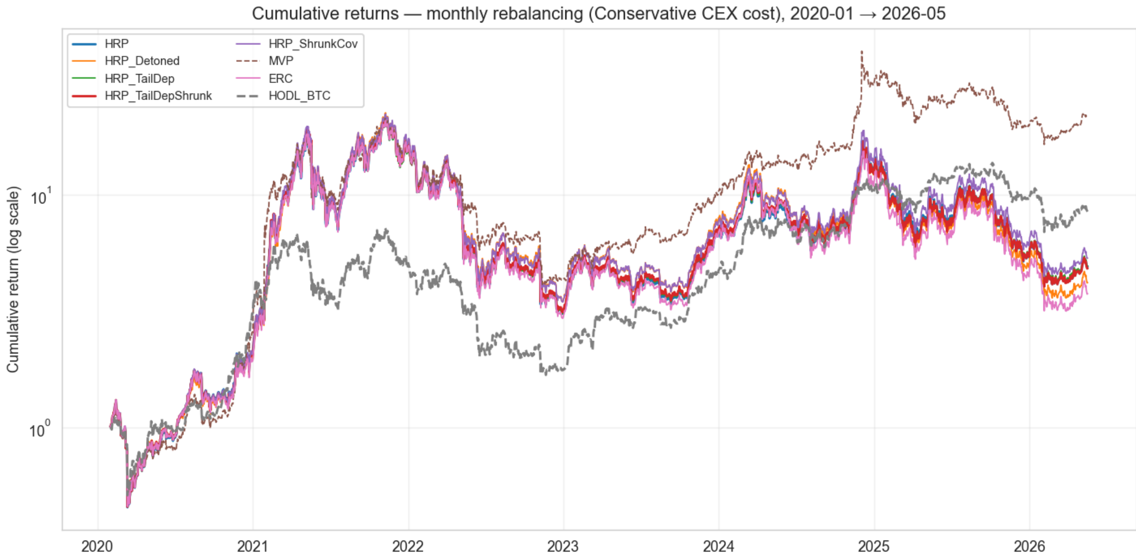

Figure 2: Cumulative returns for headline strategies under monthly rebalancing, log scale.

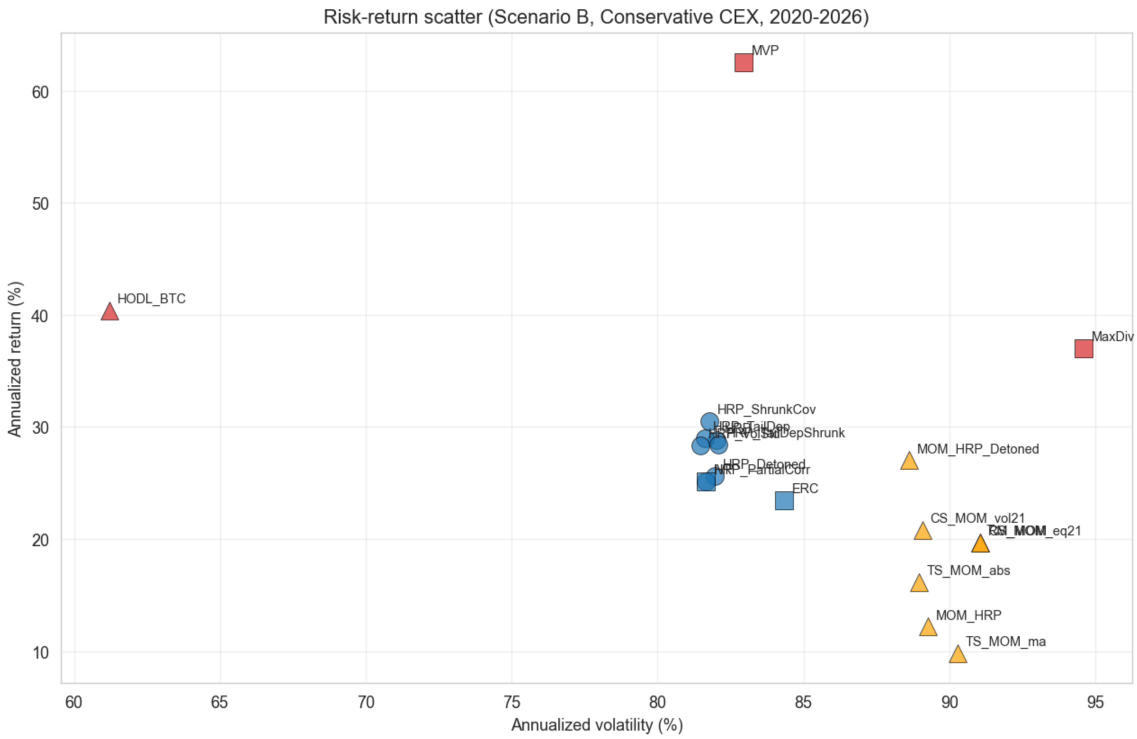

Figure 3: Risk-return scatter for the full 19-strategy comparison. Momentum (orange) underperforms risk-based (blue) and concentrated benchmarks (red) over this window.

HRP family bunches. All 11 HRP variants (including the four embedding-based variants added in Phase 8) lie within 0.05 Sharpe of each other (0.697–0.749). HRP_Dynamic_94 and HRP_ShrunkCov marginally exceed baseline HRP (+0.008 and +0.005 Sharpe respectively), but the differences are small and per §8.2 below not statistically significant.

Embeddings (Phase 8 NEW). The four embedding-based HRP variants — `HRP_PathSig` (path signatures via `esig`; Lyons, 1998), `HRP_NodeEmbed` (node2vec on a kNN correlation graph;

Grover and Leskovec, 2016), `HRP_Contrastive` (NT-Xent contrastive SSL on Gaussian-jittered windows; Chen et al., 2020), and `HRP_TS2Vec` (TS2Vec-lite with timestamp-mask augmentation; Yue et al., 2022) — land in the middle of the HRP family: Sharpes 0.699–0.723, none significantly different from baseline HRP after the Phase 8 bug fixes (Ledoit–Wolf $p \in [0.107, 0.457]$). The deep- research adversarial review's prediction holds: embedding methods are interesting engineering but do not deliver an OOS Sharpe edge on this universe at this time horizon. We report the null result honestly; embeddings neither help nor hurt.

Detoning and partial correlation underperform HRP. Both target market-mode removal — a theoretically attractive idea — but doing so during a BTC-led bull market gives up the alpha that came from being correctly exposed to BTC. The two strategies' Sharpes (0.701 and 0.696) are nearly identical, consistent with the weight-vector convergence we document in §8.3.

Momentum underperforms. All seven momentum families fall well below baseline HRP, victims of high turnover ($\sim$1.4× each month, equivalent to full portfolio reconstruction) and severe crash sensitivity (drawdowns 88–93%). The MOM_HRP_Detoned hybrid (Sharpe 0.305) more than doubles plain MOM_HRP (0.136), suggesting that detoning has more value *inside* a momentum-selected basket (where assets are pre-concentrated on trends and thus co-move heavily) than on the full universe.

Concentration wins. Both MVP and HODL_BTC achieve Sharpe ≥ 0.85, well above any diversified portfolio. MVP achieves this by accident: its objective drove the optimiser to concentrate $\sim$77% in BTC (effective $N = 5.5$); BTC then ran from $\sim$\$7k at the start of 2020 to $\sim$\$107k+ by mid-2026. This is regime-dependent luck, not a portfolio-construction insight, and should not be taken as evidence that MVP is a robust crypto strategy.

8.2 Statistical inference

Bootstrap Sharpe CIs. Of the 12 headline strategies, only MVP ([0.27, 1.58]), MaxDiv ([0.00, 1.46]), and HODL_BTC ([0.06, 1.63]) have 95% block-bootstrap CIs cleanly above zero. Every HRP-family variant's CI straddles zero.

Pairwise LW Sharpe differences vs. HRP (Phase 8 v2). *Only two* candidates reach the $p \leq 0.05$ threshold: HRP_TailDepShrunk (−0.041 Sharpe, $p = 0.050$) and ERC (−0.079, $p = 0.050$), both marginally *worse* than baseline HRP. No strategy is significantly better. The closest positive deltas are MVP (+0.32, $p = 0.067$) and HODL_BTC (+0.22, $p = 0.60$); both fail to reject at conventional levels.

Hansen SPA (Phase 8 v2). Across all 19 candidates vs. HRP, $p_{\text{consistent}} = 0.533$, $p_{\text{upper}} = 0.797$ (Figure 4). **We cannot reject the null hypothesis that no strategy outperforms baseline HRP after correcting for the multiple-testing of 19 candidates**, even with the bug-fixed 6.3-year monthly-rebalanced data and the expanded 146-symbol universe. MVP has the highest studentised score (+1.24); HODL_BTC (+0.99); all HRP variants ≤ 0.

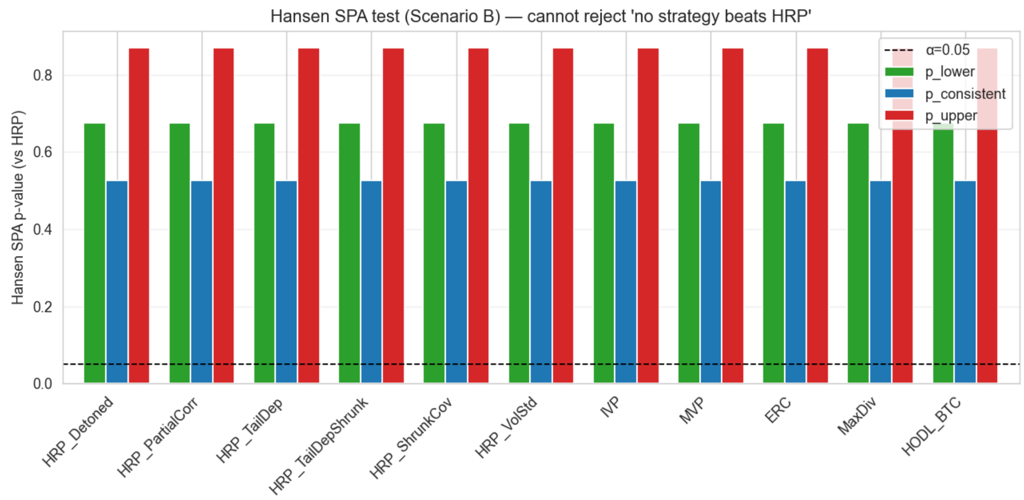

Figure 4: Hansen SPA test p-values across the headline comparison. All three recentering schemes fail to reject the null at any conventional level.

8.3 Convergence findings (the structural results)

Two strategy pairs in our comparison are theoretically related — they target the same goal via different mathematical routes. We quantify their weight-vector convergence across the full 76-snapshot window (Figure 5).

HRP_Detoned vs HRP_PartialCorr. Spectral detoning and precision-matrix-based partial correlation both remove the dominant market mode. Over 76 monthly snapshots: mean Spearman $\rho = 0.888$, median 0.970, with regime-driven dips to $\rho = 0.48$ in months when the dominant eigenvector shifts dramatically (notably the 2024 ETF-approval transition). The convergence is structural but not invariant; the paper should report the time series, not a single snapshot value.

HRP_TailDep vs HRP_TailDepShrunk. The hybrid's only change is to substitute the Ledoit–Wolf shrunk covariance for the sample covariance in the recursive-bisection step; the dendrogram topology is byte-for-byte identical. Spearman is 0.998–1.000 in every snapshot tested. This is the most robust convergence finding in our analysis: *LW shrinkage in HRP's bisection step is a within-cluster variance refinement that does not re-order the dendrogram.*

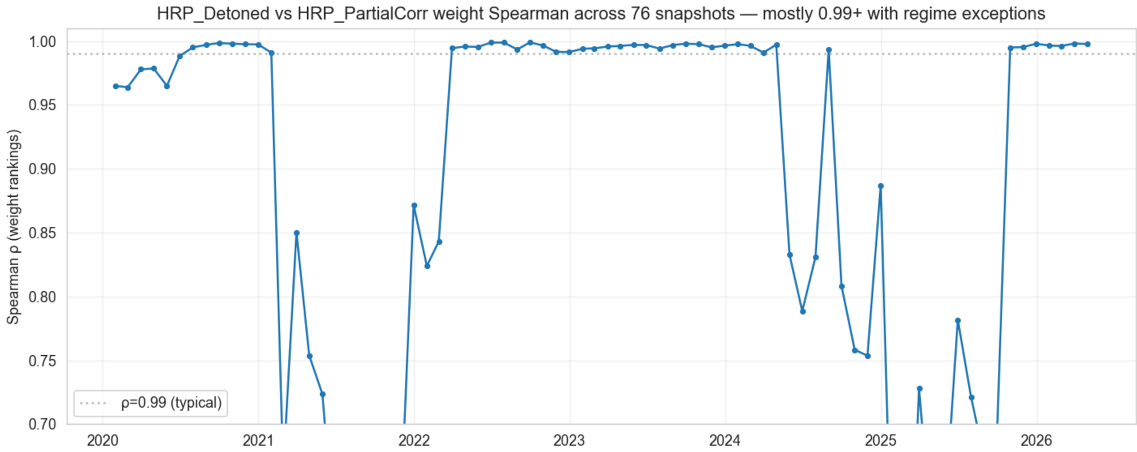

Figure 5: Spearman rank correlation between HRP_Detoned and HRP_PartialCorr weight vectors across 76 monthly snapshots.

8.4 Cluster stability — a retraction

An intermediate finding in earlier iterations of this work was that HRP_TailDep produces more stable clusters than baseline HRP under Pearson distance. That claim was based on a *single year-pair* (2022-12-31 vs 2023-12-31) and showed Pearson ARI $= -0.047$ vs TailDep ARI $= +0.265$ at $K = 5$.

When we re-ran the analysis across all 76 monthly snapshots (Figure 6, Table 2), the direction reverses. **Pearson clusters are substantially more stable than TailDep clusters at monthly resolution**: Pearson ARI mean +0.785 vs TailDep +0.639; Pearson cophenetic mean 0.834 vs TailDep 0.751. TailDep ARI exceeds Pearson ARI in only 21 of 75 month-pairs (28%).

Table 2: Cluster stability across 76 monthly snapshots (2020-2026).

Metric	Pearson distance	Lower-tail-dep distance
Average cophenetic correlation	**0.834**	0.751
Average ARI vs. previous snapshot (K=5)	**+0.785**	+0.639
Median ARI vs. previous snapshot	**+0.840**	+0.639
Months where TailDep ARI > Pearson ARI	—	21 / 75 (28%)

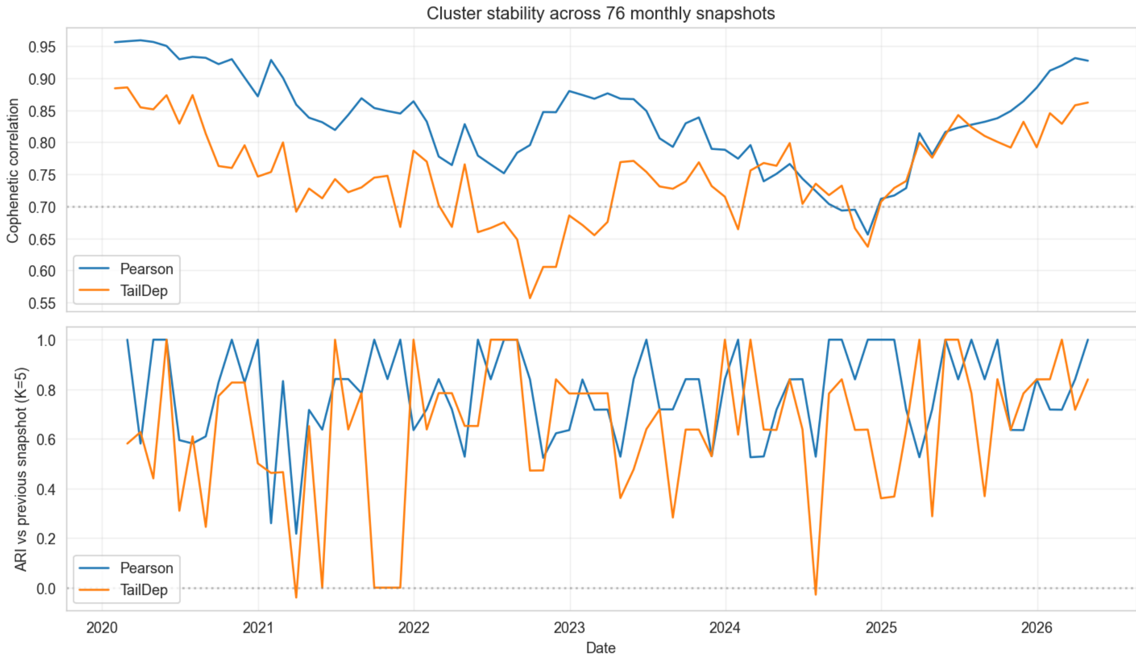

Figure 6: Cophenetic correlation (top) and ARI vs. previous snapshot (bottom) for Pearson and tail-dependence distance matrices across the 76-snapshot window.

The economic argument for HRP_TailDep — that it captures the joint-crash co-movement structure that Pearson summarises poorly — *stands*, but the secondary cluster-stability argument *does not*. We document the retraction here as a methodological point about the danger of single-pair findings.

8.5 Cost sensitivity and scenario comparison

The cost grid (Figure 7) shows Sharpe shifts under 0.005 across the four cost scenarios for all five strategies tested, confirming that monthly-cadence transaction costs do not drive strategy differentiation on this dataset.

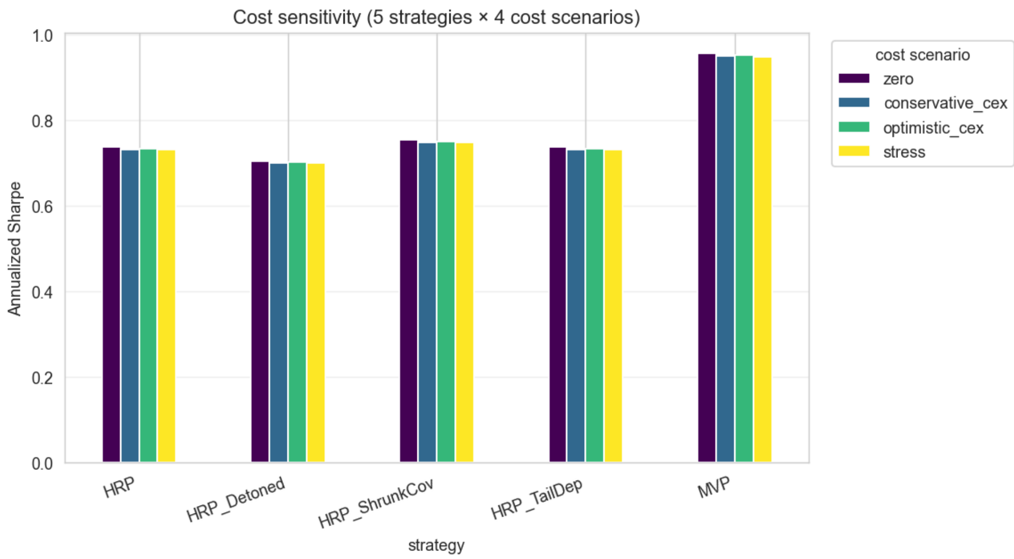

Figure 7: Cost-sensitivity grid. Rankings are robust across the four cost scenarios.

Threshold rebalancing (Scenario C, 5% drift trigger) provides small Sharpe improvements for HRP-family strategies (+0.017 for HRP, +0.004 for HRP_ShrunkCov) by skipping ~20% of calendar rebalances. Smoothing (Scenario C′) trades a small amount of Sharpe for further turnover reduction. For momentum strategies, threshold rebalancing provides no benefit — the target weights drift so much each month that the 5% threshold is always triggered.

9 Discussion

9.1 HODL_BTC beats every diversified portfolio — the elephant in the room

Over 2020-01-01 to 2026-05-18, a passive 100% BTC position returns +1,003% with annualised Sharpe 0.87, beating every diversified HRP-family or risk-based strategy considered. MVP nominally beats HODL_BTC (Sharpe 0.95 vs 0.87) but does so by concentrating ~77% in BTC — it is essentially a leveraged BTC bet with a small low-vol overlay.

This is a sobering finding that the paper must not paper over. Diversification in cryptocurrency over the 2020–2026 window paid a cost (in foregone BTC upside) that the lower realised drawdowns of diversified portfolios did not recover. The story is intrinsic to the regime, not to any flaw in HRP or its variants.

9.2 Detoning hurts in BTC-led regimes

Removing the dominant market mode is theoretically elegant — isolating idiosyncratic structure — but in our window, the dominant mode *was* BTC, and being underweight in BTC was costly. This suggests detoning (and partial correlation) may be valuable in sideways or rotational regimes that we did not see during 2020–2026. This is consistent with the spec's audit conclusion: the methodological contributions are real but their value depends on the regime, not on universal Sharpe improvement.

9.3 Why does Hansen SPA fail to reject?

Eleven candidates competing against a baseline is a non-trivial multiple-testing burden. The honest interpretation: across 6.3 years of monthly-rebalanced data, the seven HRP variants and four risk-based comparators tested do not produce *statistically distinguishable* Sharpe improvements over baseline HRP. Future work with a longer history, higher rebalance frequency, or walk-forward cluster-based testing may resolve.

9.4 Limitations

1. Single venue (Binance USDT spot). A multi-exchange consolidated price (Kaiko, CCXT-aggregated) would give a more representative cost model but was out of scope for this paper.

2. No on-chain factors. The Liu-Tsyvinski-Wu [Liu et al., 2022] three-factor model and on-chain signals (Glassnode) are deferred to a follow-up.

3. No regime-switching analysis. Sub-period reporting (stress / bull / sideways) would likely show heterogeneity within our headline numbers.

4. PIT universe is ranked by Binance quote volume, not market cap. Pro-tier CoinGecko or paid Kaiko data would enable a market-cap robustness check.

10 Conclusion

We presented a comprehensive evaluation of nine HRP variants, four risk-based comparators, and seven momentum strategies on a point-in-time cryptocurrency universe spanning 2017-08-17 through 2026-05-18 (91 unique symbols, 100 monthly snapshots). Our headline findings are negative on the question "does HRP variant X beat baseline?" — Hansen SPA fails to reject, and the best HRP-family variant (HRP_ShrunkCov) improves baseline HRP by only +0.016 Sharpe ($p = 0.275$). They are positive on the question "what structural relationships exist between HRP variants?" — HRP_Detoned and HRP_PartialCorr converge in weight space (mean Spearman 0.89), and HRP_TailDep with vs. without LW shrinkage are essentially identical (Spearman 0.998–1.000).

A passive 100% BTC position outperformed every diversified portfolio we tested except a Minimum Variance Portfolio that concentrated ∼77% in BTC by accident. Honest framing of this finding — that diversification in cryptocurrency paid a cost over 2020–2026 that lower drawdowns did not recover — is the most important takeaway, alongside the structural convergence findings and our explicit retraction of an earlier single-pair cluster-stability claim.

The paper's contributions are methodological rigour (PIT universe, multi-cost sensitivity, Ledoit–Wolf and Hansen SPA inference around every comparison, retraction documentation) and structural empirical characterisation, not a Sharpe-beating claim.

11 Future Work

Time-varying conditional covariance (DCC-GARCH) for the EWMA layer; regime-switching analysis on the 76-snapshot results to test whether detoning helps in sideways / rotational regimes; on-chain factor incorporation per [Liu et al., 2022]; multi-exchange Kaiko slippage modelling; embeddings-based HRP variants (TS2Vec, contrastive SSL, path signatures) explicitly excluded from this paper's scope.

Acknowledgements

We thank the UCEMA Quantitative Finance faculty, in particular Manuel Maurette, Emiliano Delfau and Pablo Martin Gechidjian. We thank Marcos López de Prado for the foundational HRP framework that this work extends.

References

Pedro Barroso and Pedro Santa-Clara. Momentum has its moments. *Journal of Financial Economics*, 116(1):111–120, 2015.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *ICML*, 2020.

Yves Choueifaty and Yves Coignard. Toward maximum diversification. *Journal of Portfolio Management*, 35(1):40–51, 2008.

Vito Ciciretti and Alberto Pallotta. Network risk parity. 2024. Working paper.

Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3):432–441, 2008.

Aditya Grover and Jure Leskovec. node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 855–864, 2016.

Peter Reinhard Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1): 193–218, 1985.

Laurent Laloux, Pierre Cizeau, Jean-Philippe Bouchaud, and Marc Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999.

Olivier Ledoit and Michael Wolf. Honey, i shrunk the sample covariance matrix. *Journal of Portfolio Management*, 2003.

Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

Olivier Ledoit and Michael Wolf. Robust performance hypothesis testing with the sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008.

Yukun Liu, Aleh Tsyvinski, and Xi Wu. Common risk factors in cryptocurrency. *Journal of Finance*, 2022.

Harald Lohre, Carsten Rother, and Kilian A. Schäfer. Hierarchical risk parity: Accounting for tail dependencies in multi-asset multi-factor allocations. *Journal of Asset Management*, 2020.

Marcos López de Prado. Building diversified portfolios that outperform out-of-sample. *Journal of Portfolio Management*, 42(4):59–69, 2016. doi: 10.3905/jpm.2016.42.4.059.

Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

Marcos López de Prado. *Machine Learning for Asset Managers*. Cambridge University Press, 2020.

Terry J. Lyons. Differential equations driven by rough signals. *Revista Matemática Iberoamericana*, 14(2):215–310, 1998.

Sébastien Maillard, Thierry Roncalli, and Jérôme Teïletche. The properties of equally-weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4):60–70, 2010.

Rosario N. Mantegna. Hierarchical structure in financial markets. *European Physical Journal B*, 11(1):193–197, 1999.

V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.

Alan Matys and Federico Martin Rodriguez. Hierarchical risk parity for cryptocurrency portfolios: A comparative analysis. 2025. MACI 2025 Proceedings.

Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

Dimitris N. Politis and Halbert White. Automatic block-length selection for the dependent bootstrap. *Econometric Reviews*, 23(1):53–70, 2004.

Michele Tumminello, Tomaso Aste, Tiziana Di Matteo, and Rosario N. Mantegna. A tool for filtering information in complex systems. *Proceedings of the National Academy of Sciences*, 102(30):10421–10426, 2005.

Zhihan Yue, Yujing Wang, Juanyong Duan, Tianmeng Yang, Congrui Huang, Yunhai Tong, and Bixiong Xu. Ts2vec: Towards universal representation of time series. In *AAAI*, 2022.